QALA

Solution Factory Operating System

Platform Requirements Document v1.0

Field	Value
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System	Qala Solution Factory Operating System
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Status	Draft — For Review
Requirement Framework	MoSCoW (Must / Should / Could / Won't)
Traceability	Requirements traced to SDD v2.0 and Workflows & Use Cases v1.0
Audience	Engineering, Architecture, Product, QA, Security, Compliance
Classification	Confidential

1 Introduction

This document defines the complete set of functional and non-functional requirements for the Qala Solution Factory Operating System (SF-OS). Requirements are expressed using the MoSCoW framework and are traceable to the Qala SDD v2.0 and Qala Workflows & Use Cases documents.

Qala is an end-to-end platform that governs the creation, management, and lifecycle of solutions of any type — applications, systems, goods, products, services, and platforms — through a structured hierarchy of Solution Development Environments (SDEs) and Solution Factories (SFs). This document captures what the platform must, should, could, and will not do in its initial release.

1.1 Document Conventions

Priority Key
M — Must Have **S — Should Have** **C — Could Have** **W — Won't Have (this release)**

Requirement IDs follow the pattern: [Domain]-[Sequence], e.g. FR-SDE-001 (Functional Requirement, SDE domain, sequence 001). Non-functional requirements use NFR-[Category]-[Sequence].

1.2 Scope

This document covers requirements across the following platform domains:

- Solution Structure Model — the hierarchical data model governing all solutions.
- Solution Development Environment (SDE) — lifecycle, governance, backup, and recovery.
- Solution Factory (SF) — factory creation, orchestration, and governance.
- Hermetic Build System — reproducible, isolated, attested builds.
- Continuous Integration (CI) — automated build, test, and validation.
- Continuous Delivery & Deployment (CD) — staged release, deployment strategies, and rollback.
- Configuration Management (CM) — versioning, change control, drift detection.
- Version Control Management — all artefact types, branching, and merge governance.
- Bug Tracking & Defect Management — defect lifecycle from discovery to closure.
- Test Management & Prototyping — test planning, execution, analytics, and sandbox management.
- Release Management — release records, gate evaluation, and deployment orchestration.
- Tooling Management — tool registry, toolchain governance, and vendor integrations.
- Security, Privacy & Threat Management — SEM, vulnerability management, and compliance.
- Benchmarking & Analytics — performance, quality, and process metrics.
- AI & Intelligence — recommendations, predictions, and automation.
- Project, Product & Resource Management — entity lifecycle and resource governance.
- Artefact & Package Management — registry, dependency, and binary management.
- Environment Management — provisioning, drift detection, and workflow automation.
- Observability — logging, metrics, tracing, and alerting.
- Platform Administration — user management, RBAC, audit, and compliance reporting.

1.3 Stakeholders

Stakeholder	Role in Requirements
Developer	Primary end-user; contributor to functional requirements.
Tech Lead	Architecture owner; approves technical constraint requirements.
QA Engineer	Owner of testing and quality gate requirements.
Release Manager	Owner of release management and deployment requirements.
Security Engineer	Owner of security, privacy, and compliance requirements.
Platform Admin	Owner of administration, governance, and policy requirements.
Product Manager	Prioritises requirements; owner of product and solution management requirements.
AI/ML Engineer	Contributor to AI capability requirements.
Compliance Officer	Owner of regulatory and audit requirements.
End User (Industry)	Represented via use cases in the Workflows & Use Cases document.

2 Solution Structure Model Requirements

Requirements governing the hierarchical data model that all solutions must conform to.

ID	Priority	Requirement Statement
FR-SM-001	M	The platform shall support six first-class solution types: Application, System, Good, Product, Service, and Platform.
FR-SM-002	M	The platform shall enforce a seven-level containment hierarchy: System > Application > Process > Component > Interface > Message > Data Structure.
FR-SM-003	M	Every Component shall expose its capabilities and dependencies exclusively through typed Interfaces.
FR-SM-004	M	Each Interface shall declare Imports (inbound) and Exports (outbound) as lists of Message references.
FR-SM-005	M	Every Message shall have a type of either Event (dynamic) or State (static).
FR-SM-006	M	Every Data Structure shall be composed of typed fields using the platform-defined type system.
FR-SM-007	M	The platform type system shall support: bool, string, char, varchar, int, float, double, date, array, tuple, set, map, object, pointer, custom, and null.
FR-SM-008	M	The platform shall enforce referential integrity within the solution hierarchy; no orphaned elements shall exist.
FR-SM-009	S	The platform shall provide a visual graph representation of the solution hierarchy for any given solution.
FR-SM-010	S	The platform shall validate that all Interface Imports are satisfied by corresponding Exports elsewhere in the solution graph.
FR-SM-011	C	The platform shall allow solution hierarchy templates to be defined and reused across SDEs and factories.
FR-SM-012	C	The platform shall support cross-solution dependency declarations between separate Solution records.

3 Solution Development Environment (SDE) Requirements

Requirements for SDE provisioning, configuration, lifecycle, governance, backup, and recovery.

3.1 Provisioning & Configuration

ID	Priority	Requirement Statement
FR-SDE-001	M	The platform shall allow any authenticated user with the Developer role to provision a new SDE via the API or portal.
FR-SDE-002	M	Every SDE shall be provisioned from either a platform-approved template or a custom configuration supplied by the requester.
FR-SDE-003	M	An SDE shall be deployable to at least three target classes: a cloud platform, a logical environment designation, and a physical or virtual machine.
FR-SDE-004	M	An SDE shall support the following configuration elements: environment variables, IDE configurations, language runtimes, user preferences, profiles, parameters, options, toolkits, and a content management system.
FR-SDE-005	M	All SDE configuration shall be version-controlled; every change shall produce a new configuration version with author and timestamp.
FR-SDE-006	M	A single SDE shall be able to host and manage multiple solutions simultaneously.
FR-SDE-007	S	The platform shall provision a new SDE within 10 minutes for any supported cloud target.
FR-SDE-008	S	The platform shall support cloning an existing SDE to a new independent instance from any named snapshot version.
FR-SDE-009	S	The platform shall enforce that all build dependencies within an SDE are pinned to exact versions with cryptographic digest verification.
FR-SDE-010	C	The platform shall support ephemeral SDEs that are automatically deprovisioned after a configurable time-to-live.

3.2 Lifecycle Management

ID	Priority	Requirement Statement
FR-SDE-011	M	Every SDE shall be governed by a defined lifecycle with the following states: Provisioning, Active, Snapshotted, Suspended, Rolled Back, Quarantined, Archived, and Decommissioned.
FR-SDE-012	M	The platform shall enforce valid state transitions; invalid transitions shall be rejected with a descriptive error.
FR-SDE-013	M	The platform shall allow any authorised user to take a point-in-time snapshot of an Active SDE at any time.
FR-SDE-014	M	A snapshot shall capture: filesystem state, configuration version, environment variable set, toolchain manifest, active solution associations, and a SHA-256 digest.

FR-SDE-015	M	Snapshot completion shall take no longer than 60 seconds for a standard-sized SDE.
FR-SDE-016	M	The platform shall allow rollback of an SDE to any stored snapshot version; rollback shall complete within 5 minutes.
FR-SDE-017	M	Prior to any rollback, the platform shall automatically take a safety snapshot of the current state.
FR-SDE-018	M	Transitions to Archived or Decommissioned states shall require approval from the SDE owner and a Platform Admin.
FR-SDE-019	S	The platform shall verify that no active pipelines, open Critical/High bugs, or pending releases are linked to an SDE before allowing decommission.
FR-SDE-020	S	Every SDE lifecycle event shall be published as an SDE_EVENT to the platform event stream.

3.3 Backup & Recovery

ID	Priority	Requirement Statement
FR-SDE-021	M	The platform shall support scheduled full and incremental backups of all Active SDEs.
FR-SDE-022	M	Every backup shall record: Backup ID, SDE ID, owner ID, environment version, timestamp, backup type, storage location, SHA-256 digest, and status.
FR-SDE-023	M	The platform shall support point-in-time restore of any SDE from any stored backup.
FR-SDE-024	M	All backups shall be encrypted at rest. Backup integrity shall be verified via digest comparison before any restore.
FR-SDE-025	M	The platform shall alert the SDE owner and Platform Admin within 15 minutes of a failed backup.
FR-SDE-026	S	The platform shall achieve a scheduled backup success rate of 99.9% or higher.
FR-SDE-027	S	The platform shall support both local and cloud-hosted backup storage targets.
FR-SDE-028	C	The AI Agent shall recommend optimal backup frequency and storage tier based on SDE activity patterns.

3.4 SDE Governance

ID	Priority	Requirement Statement
FR-SDE-029	M	Every SDE shall have a designated owner. Ownership transfers shall be recorded in the audit trail.
FR-SDE-030	M	All SDE configuration changes shall be tracked with: actor, timestamp, change description, and before/after diff.
FR-SDE-031	M	Platform-wide security and compliance policies shall be continuously evaluated against all Active SDEs.

FR-SDE-032	M	Policy violations shall generate an alert and a remediation checklist for the SDE owner within a configurable SLA window.
FR-SDE-033	S	The platform shall support SDE archival with configurable retention policies and compliant long-term storage.
FR-SDE-034	S	Archived SDEs shall be retrievable and reactivatable with full dependency restoration.
FR-SDE-035	C	The platform shall detect inactive SDEs exceeding a configurable threshold and recommend archival to the Platform Admin.

4 Solution Factory (SF) Requirements

Requirements for factory creation, SDE membership, orchestration, and factory-wide governance.

ID	Priority	Requirement Statement
FR-SF-001	M	The platform shall allow a Platform Admin to create a Solution Factory by grouping two or more SDEs under shared governance.
FR-SF-002	M	Any Active, non-quarantined SDE shall be enrollable into a Solution Factory.
FR-SF-003	M	A single SDE shall be permitted to belong to multiple Solution Factories simultaneously.
FR-SF-004	M	The Solution Factory shall enforce platform-wide policies (security, tooling, coding standards) against all enrolled SDEs at enrolment and continuously thereafter.
FR-SF-005	M	Non-compliant SDEs shall receive a remediation checklist and a configurable grace period before enrolment is blocked.
FR-SF-006	M	The Solution Factory shall provide a unified operational view across all enrolled SDEs: aggregate event stream, health status, and shared artefact registry.
FR-SF-007	M	Factory-level security policy updates shall be propagated to all enrolled SDEs with a dry-run preview before live activation.
FR-SF-008	S	The Solution Factory shall support cross-SDE artefact sharing with access control governed at the factory level.
FR-SF-009	S	The platform shall provide factory-level analytics comparing quality, performance, and process metrics across all enrolled SDEs.
FR-SF-010	C	The platform shall support hierarchical factory structures where one factory can be a member of a higher-level factory.
FR-SF-011	C	The AI Agent shall identify high-performing patterns across factory members and surface them as factory-wide recommendations.

5 Hermetic Build System Requirements

Requirements for reproducible, isolated, attested build environments.

ID	Priority	Requirement Statement
FR-HB-001	M	Every build shall execute within a fully isolated container with no access to host network, filesystem, or ambient credentials beyond explicitly declared inputs.
FR-HB-002	M	All build environment definitions shall be stored in version control and treated as immutable once activated.
FR-HB-003	M	Every declared build dependency shall be pinned to an exact version with a SHA-256 cryptographic digest. References to floating version ranges (e.g. "latest") shall be rejected.
FR-HB-004	M	The platform shall verify the SHA-256 digest of every fetched dependency before use. A digest mismatch shall immediately fail the build and publish a SECURITY_EVENT.
FR-HB-005	M	Given identical source commit and environment definition, the platform shall produce bit-for-bit identical build outputs across repeated executions.
FR-HB-006	M	Every successful build shall produce a signed build attestation record containing: build_id, source_commit SHA, env_definition_hash, builder_identity, inputs_hash, outputs_hash, timestamp, and cryptographic signature.
FR-HB-007	M	Build attestation shall be stored in the Artefact Registry alongside the produced artefact and must be verifiable at all downstream promotion gates.
FR-HB-008	M	Any artefact lacking a valid, verifiable attestation shall be blocked from promotion beyond the build stage.
FR-HB-009	M	Build environment definition changes shall require a Configuration Change Request (CCR) approved before activation.
FR-HB-010	S	The platform shall target SLSA Level 3 compliance for all builds: hermetic environment, all inputs tracked, no persistent credential access during build.
FR-HB-011	S	The platform shall provide a build environment validation pipeline that tests a new definition against representative builds before activation.
FR-HB-012	C	The platform shall support SLSA Level 4 compliance: two-party review of all build process changes and platform-wide reproducibility enforcement.

6 Continuous Integration (CI) Requirements

Requirements for automated build, test, analysis, and validation on every code change.

ID	Priority	Requirement Statement
FR-CI-001	M	The platform shall automatically trigger a CI pipeline on every commit to a version-controlled branch.
FR-CI-002	M	Every CI pipeline shall execute the following stages in order: source validation, hermetic environment provisioning, dependency fetch and digest verification, SAST, build, unit tests, SCA, and artefact packaging.
FR-CI-003	M	Integration test execution shall be supported as a CI stage against automatically provisioned ephemeral test environments.
FR-CI-004	M	Each CI stage shall be independently configurable as a pass/fail gate; a failing gate shall halt the pipeline and prevent artefact publication.
FR-CI-005	M	Pipeline results, including full logs, shall be published as BUILD_EVENTS to the platform event stream and accessible via the portal within 2 minutes of completion.
FR-CI-006	M	A build with any SAST finding at or above the configured severity threshold shall fail the pipeline.
FR-CI-007	M	Unit test code coverage below the configured minimum threshold shall fail the CI pipeline.
FR-CI-008	M	Discovery of any Critical or High severity CVE by SCA shall fail the CI pipeline and automatically create a security bug record.
FR-CI-009	M	Container image builds shall be scanned for OS-level vulnerabilities as a mandatory CI stage.
FR-CI-010	S	The platform shall support parallel execution of independent CI stages to minimise pipeline duration.
FR-CI-011	S	The platform shall track pipeline duration trends and alert when any stage duration increases by more than 20% relative to its 14-day moving average.
FR-CI-012	S	The AI Agent shall prioritise test execution order within CI based on historical failure rate and code change proximity.
FR-CI-013	C	The platform shall support Infrastructure-as-Code (IaC) scanning as a CI stage for Terraform and Helm chart changes.

7 Continuous Delivery, Release & Deployment Requirements

Requirements for release planning, gate evaluation, deployment strategies, and rollback.

7.1 Release Management

ID	Priority	Requirement Statement
FR-REL-001	M	The platform shall support creation of a Release record containing: version, type, target environment, deployment strategy, linked artefacts, linked CCRs, linked bug fixes, and gate checklist.
FR-REL-002	M	All artefacts linked to a release shall be verified to carry valid build attestations before the release enters gate evaluation.
FR-REL-003	M	The platform shall enforce the following mandatory release gates: Test Gate, Coverage Gate, Benchmark Gate, Security Gate, Bug Gate, Attestation Gate, Change Control Gate, and Approval Gate.
FR-REL-004	M	Any failed mandatory gate shall block deployment and publish a RELEASE_EVENT (GATE_FAILED) to all stakeholders.
FR-REL-005	M	Critical or High severity open bugs in the release scope shall fail the Bug Gate unless a documented risk acceptance sign-off is recorded.
FR-REL-006	M	All configuration changes in the release scope shall reference approved CCRs; any unapproved change shall fail the Change Control Gate.
FR-REL-007	S	The platform shall support configurable additional gates per release type or target environment.
FR-REL-008	S	The Approval Gate shall require sign-off from a configurable list of named approvers and roles.

7.2 Deployment

ID	Priority	Requirement Statement
FR-DEP-001	M	The platform shall support Blue/Green deployment: two environments maintained, traffic switched atomically on approval.
FR-DEP-002	M	Blue/Green deployment shall support instant rollback by reversing the traffic switch without redeployment.
FR-DEP-003	M	The platform shall support Canary deployment: new version receives a configurable initial traffic percentage, incrementing on configurable observation windows and health thresholds.
FR-DEP-004	M	Canary deployment shall support automatic rollback when any monitored metric breaches a configurable threshold during an observation window.
FR-DEP-005	M	The platform shall support Rolling deployment across instance fleets with configurable batch size and health check interval.
FR-DEP-006	M	The platform shall support Feature Flag-based deployment: code deployed dark and enabled selectively via configuration without redeployment.

FR-DEP-007	M	All deployment events shall be published to the BUILD_EVENTS and NOTIFICATIONS Kafka topics in real time.
FR-DEP-008	M	A post-deployment rollback shall be executable by an authorised Release Manager within 5 minutes of identifying an issue.
FR-DEP-009	S	The platform shall support a configurable manual hold stage in Blue/Green deployments between provisioning and traffic switch.
FR-DEP-010	S	Smoke tests shall run automatically after every deployment, before any user traffic is admitted.
FR-DEP-011	C	The platform shall support deployment to multi-cloud targets from a single release record.

8 Configuration Management & Version Control Requirements

Requirements for CCR governance, configuration versioning, drift detection, and version control of all platform artefact types.

8.1 Change Control

ID	Priority	Requirement Statement
FR-CM-001	M	The platform shall support a Configuration Change Request (CCR) workflow for all changes to: SDE configuration, build environment definitions, service configuration, infrastructure definitions, security policies, and tool versions.
FR-CM-002	M	Every CCR shall record: ID, title, requester, change type, affected scope, risk assessment, justification, implementation plan, rollback plan, and approval status.
FR-CM-003	M	The platform shall auto-score each CCR for risk based on change type, scope breadth, and historical change failure rate.
FR-CM-004	M	High-risk CCRs shall be routed to a Change Control Board (CCB) for review. Low-risk CCRs may be approved by a single designated approver.
FR-CM-005	M	An Emergency Change process shall be supported: single emergency approver; mandatory post-implementation CCB review within 24 hours.
FR-CM-006	M	All CCR state transitions shall be published as CM_EVENTS and immutably recorded in the audit trail.
FR-CM-007	S	The platform shall enforce a post-change validation run (automated tests, scans, health checks) after every approved CCR implementation.
FR-CM-008	S	The platform shall block deployment of any configuration change not covered by an approved CCR.

8.2 Configuration Versioning & Drift Detection

ID	Priority	Requirement Statement
FR-CM-009	M	All configuration artefact types shall be stored in version control with full history: environment variables, build definitions, service configs, IaC, security policies, and tool manifests.
FR-CM-010	M	The platform shall continuously monitor all Active SDEs and environments for configuration drift from their version-controlled definitions.
FR-CM-011	M	Any detected drift shall generate an alert to the SDE owner and Platform Admin within 15 minutes of detection.
FR-CM-012	M	Drift resolution shall result in either: remediation (environment reverted to definition) or formalisation (new CCR raised to make the drift the new definition).
FR-CM-013	S	Rollback to any prior configuration version shall be supported for all versioned configuration artefact types.
FR-CM-014	S	The platform shall maintain a complete version history for all tool versions used, including deprecated versions accessible for historical reproducibility.

9 Bug Tracking & Defect Management Requirements

Requirements for defect recording, triage, lifecycle, and integration with CI/CD and release gates.

ID	Priority	Requirement Statement
FR-BUG-001	M	The platform shall provide a defect tracking system supporting creation, triage, assignment, resolution, verification, and closure of bug records.
FR-BUG-002	M	Every bug record shall capture: ID, title, severity (Critical/High/Medium/Low), priority (P1–P4), status, reporter, assignee, environment, linked release version, linked feature, steps to reproduce, and MTTR.
FR-BUG-003	M	Bug records shall be auto-created by: failed automated test cases, security scan findings, and production monitoring alerts.
FR-BUG-004	M	Bug records created from automated test failures shall be automatically linked to the failing test case, the build ID, and the offending commit SHA.
FR-BUG-005	M	The platform shall prevent any release from being promoted to production with open Critical or High severity bugs in scope, unless a documented risk acceptance sign-off is recorded.
FR-BUG-006	M	A fix commit shall be linkable to one or more bug records. Regression test cases shall be linkable to the bug records they protect.
FR-BUG-007	M	All bug lifecycle transitions shall be published as BUG_EVENTS to the platform event stream.
FR-BUG-008	S	The platform shall detect regression: a CI build that causes a previously passing test to fail shall auto-create a bug record linked to the offending commit.
FR-BUG-009	S	The AI Agent shall analyse new bug reports against historical defect patterns and suggest likely root cause area, similar past bugs, and relevant test cases.
FR-BUG-010	S	Defect density and MTTR metrics shall be computed per project, application, and release and surfaced on the quality benchmarking dashboard.
FR-BUG-011	C	The platform shall support a "Won't Fix" disposition with mandatory justification and risk acceptance record.

10 Test Management & Prototyping Requirements

Requirements for test planning, test execution, test environment management, analytics, and sandbox prototyping.

10.1 Test Management

ID	Priority	Requirement Statement
FR-TEST-001	M	The platform shall provide a Test Management Service supporting: test case creation, test plan definition, test suite organisation, test run execution, and result recording.
FR-TEST-002	M	Test cases shall be linkable to: features, user stories, bug records, and release records.
FR-TEST-003	M	Test cases shall support the following types: Unit, Integration, System, Acceptance, Security, Performance, and Regression.
FR-TEST-004	M	Test environments shall be automatically provisioned from version-controlled, approved templates on test run trigger.
FR-TEST-005	M	Ephemeral test environments shall be torn down automatically on test run completion unless explicitly retained.
FR-TEST-006	M	Test results shall be published as TEST_EVENTS and linked to the associated build, pipeline, and release records.
FR-TEST-007	M	Failed test cases shall automatically create or update linked bug records.
FR-TEST-008	S	The platform shall detect flaky tests: tests that non-deterministically pass or fail across runs. Flaky tests shall be flagged and routed to a dedicated review queue.
FR-TEST-009	S	The AI Agent shall generate test cases targeting code paths modified in a linked commit set, presented to QA for review and acceptance.
FR-TEST-010	S	The AI Agent shall maintain a self-healing capability: automatically detecting and correcting stale test selectors and assertions in dynamic UI/API environments.
FR-TEST-011	S	Test analytics shall provide: pass rate trends, coverage trends, flakiness rates, and MTTR for test failures across all projects.
FR-TEST-012	C	The AI Agent shall prioritise which tests to execute first based on risk score and historical failure rate for a given change set.

10.2 Prototyping

ID	Priority	Requirement Statement
FR-PROTO-001	M	The platform shall provide a Prototyping Service that provisions time-bounded, resource-capped sandbox SDEs for POC and prototype work.
FR-PROTO-002	M	Prototype SDEs shall be isolated from production systems and data.
FR-PROTO-003	M	Prototype outcomes (performance telemetry, user feedback, defect discoveries) shall be recordable and linkable to product feature records.

FR-PROTO-004	S	A completed prototype SDE shall be promotable to a full developer SDE via the clone workflow, carrying all configuration and content.
FR-PROTO-005	S	The AI Agent shall generate a structured feedback and insight report from prototype telemetry data.
FR-PROTO-006	C	Prototype resource consumption shall be metered and reportable per project for cost allocation.

11 Tooling Management Requirements

Requirements for the tool registry, toolchain governance, tool version control, and vendor integrations.

ID	Priority	Requirement Statement
FR-TOOL-001	M	The platform shall maintain a Tool Registry storing: name, version, binary digest (SHA-256), source or vendor, licence identifier, provenance attestation, and approved SDE profile associations for every tool.
FR-TOOL-002	M	Tools shall be organised into a three-level hierarchy: Toolkit > Toolset > Tool.
FR-TOOL-003	M	Tools within a Toolset may be linked in dependency sequences to form Toolchains.
FR-TOOL-004	M	Every tool upgrade shall require a CCR approved before the new version is activated in any SDE or pipeline.
FR-TOOL-005	M	Multiple versions of a tool shall coexist in the registry; SDEs shall reference specific versions via Toolset definitions.
FR-TOOL-006	M	Deprecated tool versions shall be archived but remain accessible for historical reproducibility.
FR-TOOL-007	M	All tools shall be security-scanned (vulnerability scan, licence compliance, supply-chain provenance check) before onboarding.
FR-TOOL-008	M	Critical security findings on a tool shall block onboarding. High findings shall require documented risk acceptance.
FR-TOOL-009	S	Platform administrators shall be able to define pre-approved Tool Suites: curated tool sets approved for specific project types or compliance regimes.
FR-TOOL-010	S	Third-party vendor tools shall be onboardable via a governed Vendor Integration Management workflow: registration, security scan, compatibility test, CCR approval, activation.
FR-TOOL-011	S	The platform shall monitor all active vendor integrations for version updates, security advisories, and licence changes, alerting the Platform Admin on detection.
FR-TOOL-012	C	Toolchain definitions shall be publishable to a platform-wide Toolkit Registry and selectable by any SDE within the factory.

12 Security, Privacy & Threat Management Requirements

Requirements for SEM, vulnerability management, threat modelling, security testing, privacy enforcement, and compliance reporting.

12.1 Security Testing

ID	Priority	Requirement Statement
FR-SEC-001	M	Static Application Security Testing (SAST) shall be a mandatory stage in every CI pipeline.
FR-SEC-002	M	Software Composition Analysis (SCA) shall be a mandatory stage in every CI pipeline, scanning all dependencies for known CVEs.
FR-SEC-003	M	Dynamic Application Security Testing (DAST) shall be supported as a CD stage against staging and QA environments.
FR-SEC-004	S	Interactive Application Security Testing (IAST) shall be supported for instrumented test execution.
FR-SEC-005	S	Container image scanning shall be a mandatory stage for any build producing a container image.
FR-SEC-006	S	Infrastructure-as-Code scanning shall be supported for Terraform and Helm chart modifications.
FR-SEC-007	S	The platform shall support managed penetration test records: scope, methodology, findings, severity, remediation tracking, and re-test scheduling.

12.2 Security Event Management (SEM)

ID	Priority	Requirement Statement
FR-SEM-001	M	The SEM shall continuously monitor all Active SDEs for anomalous behaviour, known threat patterns, and policy violations.
FR-SEM-002	M	On confirmed threat detection, the SEM shall automatically quarantine the affected SDE (no ingress/egress) within 90 seconds.
FR-SEM-003	M	A quarantined SDE shall be restorable to Active only after a clean full security scan and explicit clearance by a Security Engineer.
FR-SEM-004	M	On SDE quarantine, the SEM shall evaluate all other SDEs in the same Solution Factory for indicators of compromise.
FR-SEM-005	M	All credentials, API keys, and secrets shall be stored in an encrypted Secrets Vault. No secrets shall be stored in code, logs, or plaintext configuration.
FR-SEM-006	M	Security policy definitions shall be version-controlled and require CCR approval before activation.
FR-SEM-007	M	All security-relevant actions shall be immutably logged with: actor, timestamp, event type, affected resource, and full event payload.
FR-SEM-008	S	The SEM shall assign a risk score and confidence level to all detected threats.

FR-SEM-009	S	The platform shall support STRIDE and PASTA as built-in threat modelling frameworks for solution risk assessments.
FR-SEM-010	S	Threat models shall be version-controlled and linked to their associated solution and release records.

12.3 Vulnerability Management

ID	Priority	Requirement Statement
FR-VULN-001	M	The platform shall enforce remediation SLAs: Critical vulnerabilities within 24 hours; High within 7 days; Medium within 30 days.
FR-VULN-002	M	Critical vulnerabilities in scope shall block all deployments from the affected SDE until remediated.
FR-VULN-003	M	Vulnerability records shall be linked to a bug record, a target fix version, and the responsible developer or team.
FR-VULN-004	S	The platform shall support automated patching for OS-level and dependency vulnerabilities, subject to CCR approval.
FR-VULN-005	S	Scheduled vulnerability scans shall run at least daily across all Active SDEs and deployed services.

12.4 Privacy Management

ID	Priority	Requirement Statement
FR-PRIV-001	M	The platform shall support classification of all data fields as Public, Internal, Confidential, or Restricted (PII/PHI).
FR-PRIV-002	M	PII and PHI fields shall be automatically masked or anonymised in all non-production environments.
FR-PRIV-003	M	All data shall be encrypted at rest using AES-256 and in transit using TLS 1.3 or higher.
FR-PRIV-004	M	Configurable data retention and deletion policies shall be enforceable per data classification.
FR-PRIV-005	M	All access to and modifications of sensitive data fields shall be captured in the immutable audit trail.
FR-PRIV-006	S	The platform shall generate compliance reports for: GDPR, CCPA, ISO 27001, SOC 2 Type II, and SLSA.
FR-PRIV-007	S	PII detection pipelines shall run on submitted datasets or content to flag accidental PII inclusion before publication.
FR-PRIV-008	C	The platform shall support a right-to-erasure workflow: locate, review, and delete all PII associated with a data subject across all SDEs and artefacts.

13 Benchmarking & Analytics Requirements

Requirements for performance, quality, and process benchmarking, dashboards, and AI-driven analytics.

ID	Priority	Requirement Statement
FR-BENCH-001	M	The platform shall collect and store performance benchmark metrics for every release promoted to staging or production: CPU, memory, disk, network, response time (p50/p95/p99), throughput, and error rate.
FR-BENCH-002	M	A performance benchmark gate shall be evaluatable as part of the release gate checklist; any metric regressing beyond configured tolerance shall fail the gate.
FR-BENCH-003	M	Code quality metrics shall be computed per build: cyclomatic complexity, maintainability index, duplication percentage, and linting violation count.
FR-BENCH-004	M	Test coverage metrics (unit, integration, system) shall be computed per build and compared to configured minimum thresholds.
FR-BENCH-005	M	Process metrics shall be tracked per project: deployment frequency, lead time from commit to production, change failure rate, and MTTR.
FR-BENCH-006	M	Automated alerts shall fire when any benchmark metric falls below a configured threshold.
FR-BENCH-007	S	All benchmark metrics shall be retained for a minimum of 12 months for trend analysis and predictive forecasting.
FR-BENCH-008	S	The platform shall support cross-project, cross-application, and cross-product benchmark comparisons with normalised scoring.
FR-BENCH-009	S	Role-specific dashboards shall be provided: Executive (KPIs, trends), Architect (code quality, technical debt), QA/Release (performance, stability).
FR-BENCH-010	S	The AI Agent shall identify high-performing teams and processes and surface recommendations to underperforming projects.
FR-BENCH-011	C	The platform shall support scenario benchmarking: simulated peak load or environment-specific stress scenarios configurable per release.
FR-BENCH-012	C	The AI Agent shall predict potential performance hotspots and defect-prone code areas based on historical benchmark data.

14 AI & Intelligence Requirements

Requirements for the AI Agent subsystem: recommendations, predictions, automation, and self-healing capabilities.

ID	Priority	Requirement Statement
FR-AI-001	M	The AI Agent shall analyse every new or updated SDE and publish optimisation recommendations within 30 minutes of a triggering event.
FR-AI-002	M	AI recommendations shall be surfaced in the portal and published to the AI_RECOMMENDATIONS Kafka topic.
FR-AI-003	M	Developers shall be able to Accept, Dismiss, or Defer each recommendation. Accepted recommendations shall be applied automatically where automation is available.
FR-AI-004	M	The AI Agent shall generate test cases for code changes, presenting them to QA for review. Accepted AI-generated test cases shall be added to the active test suite.
FR-AI-005	M	The AI Agent shall analyse the historical defect density of changed modules and produce a risk score per module for every commit.
FR-AI-006	M	The AI Agent shall detect flaky tests and propose or automatically apply fixes for deterministic root causes.
FR-AI-007	S	The AI Agent shall predict pipeline bottlenecks based on stage duration trends and forecast when a stage will breach its configured SLA.
FR-AI-008	S	The AI Agent shall monitor production and staging environments for anomalous behaviour and surface findings as security or performance alerts.
FR-AI-009	S	For AI/ML solutions, the AI Agent shall monitor prediction confidence distributions and alert on statistical drift indicative of model degradation.
FR-AI-010	S	The AI Agent shall analyse prototype telemetry and generate structured insight reports comparing outcomes against defined success criteria.
FR-AI-011	S	AI recommendation acceptance rate shall be tracked per user, per project, and platform-wide. The AI model shall be tunable based on acceptance/rejection feedback.
FR-AI-012	C	The AI Agent shall generate cross-product insights: identify patterns from high-performing projects and recommend their adoption elsewhere.
FR-AI-013	C	The AI Agent shall support a semantic knowledge retrieval capability: accepting natural language queries and returning relevant versioned artefacts and context.

15 Project, Product & Resource Management Requirements

Requirements for project, product, and resource lifecycle management across all solution types.

ID	Priority	Requirement Statement
FR-PM-001	M	The platform shall support project management with: epics, stories, tasks, bugs, and sub-tasks organised in a hierarchy.
FR-PM-002	M	Work items shall be linkable to: solutions, SDEs, pipelines, releases, test cases, and bug records.
FR-PM-003	M	Sprint planning shall be supported: work items assigned to sprints with velocity tracking and burndown.
FR-PM-004	M	All six solution types (Application, System, Good, Product, Service, Platform) shall have managed lifecycle records: creation, versioning, release history, and archival.
FR-PM-005	M	Products and Services shall support configurable distribution channels and access-controlled distribution lists.
FR-PM-006	M	Release packages for Products shall include: artefact manifest, attestations, release notes, migration guides, and rollback procedures.
FR-PM-007	S	Resource management shall support team assignments with workload visualisation and skill-matrix matching.
FR-PM-008	S	Cross-project dependency tracking shall identify and visualise blockers and dependencies across projects and applications.
FR-PM-009	S	All project metrics (velocity, lead time, deployment frequency) shall feed into the Benchmarking Service for cross-project analytics.
FR-PM-010	C	The platform shall support physical good solutions: production tracking, batch management, distribution records, and recall readiness.

16 Artefact, Package & Binary Management Requirements

Requirements for the artefact registry, dependency management, container management, and third-party software.

ID	Priority	Requirement Statement
FR-ART-001	M	The platform shall provide a centralised, immutable Artefact Registry for all compiled outputs: binaries, libraries, container images, packages, and configuration artefacts.
FR-ART-002	M	Every artefact shall be stored with: ID, name, version, type, SHA-256 digest, attestation reference, linked build ID, linked release IDs, dependency manifest, and SPDX licence identifiers.
FR-ART-003	M	Artefact content shall be immutable once stored; any change shall produce a new version.
FR-ART-004	M	All dependencies shall be declared in lock files with exact version pinning and digest verification.
FR-ART-005	M	Transitive dependencies shall be fully resolved and recorded in the dependency manifest at build time.
FR-ART-006	M	Dependency vulnerability scanning shall run against the registry on a scheduled daily basis in addition to per-build scanning.
FR-ART-007	M	All container images shall be signed using the platform signing key and verified at deployment.
FR-ART-008	M	Licence compliance checks shall flag any artefact with a dependency whose licence is incompatible with the project's declared licence policy.
FR-ART-009	S	Base image updates shall automatically trigger rebuild and retest pipelines for all services dependent on that base image.
FR-ART-010	S	Artefact retention policies shall be configurable per artefact type; artefacts past retention shall be archived to cold storage.
FR-ART-011	S	Recalled artefacts shall be immediately flagged in all linked release and deployment records, and all consumers notified.
FR-ART-012	C	The platform shall support a Software Bill of Materials (SBOM) export in SPDX or CycloneDX format for any artefact.

17 Environment Management Requirements

Requirements for environment definition, provisioning, drift detection, and workflow automation.

ID	Priority	Requirement Statement
FR-ENV-001	M	All environments shall be defined as code in version-controlled configuration files specifying: base image, OS packages, middleware, databases, network topology, security hardening, and monitoring.
FR-ENV-002	M	Environment definitions shall be subject to CCR approval before activation.
FR-ENV-003	M	The platform shall support automated provisioning of environments via containerisation (Docker) and orchestration (Kubernetes/Helm) or cloud-native IaC (Terraform/Pulumi).
FR-ENV-004	M	Ephemeral environments shall be automatically provisioned on: CI pipeline trigger, pull request creation, test run initiation, and prototype request.
FR-ENV-005	M	Ephemeral environments shall have configurable TTLs and shall be automatically deprovisioned on expiry or pipeline completion.
FR-ENV-006	M	The platform shall support cloning a production environment to create a staging mirror on demand.
FR-ENV-007	M	Continuous drift detection shall compare all running environments against their version-controlled definitions and alert on any deviation.
FR-ENV-008	S	Pre-configured project templates shall include full pipeline definitions: build, test, review, merge, and deploy workflows.
FR-ENV-009	S	The platform shall support environment snapshot and rollback consistent with the SDE snapshot/rollback workflow.
FR-ENV-010	C	IDE plugins shall allow developers to trigger environment operations (snapshot, deploy, scan) directly from their development IDE.

18 Observability Requirements

Requirements for logging, metrics, distributed tracing, and alerting across all platform services.

ID	Priority	Requirement Statement
FR-OBS-001	M	All platform services shall emit structured JSON logs including: service name, trace ID, span ID, timestamp, log level (DEBUG/INFO/WARN/ERROR/FATAL), and message.
FR-OBS-002	M	All logs shall be forwarded to a centralised log aggregation system and retained for a minimum of 90 days in hot storage.
FR-OBS-003	M	All services shall emit service-level metrics to a centralised metrics store: request rate, error rate, latency (p50/p95/p99), and saturation.
FR-OBS-004	M	All inter-service calls shall be instrumented with distributed tracing (OpenTelemetry compatible). Trace context shall be propagated through all Kafka event payloads.
FR-OBS-005	M	Configurable alerting rules shall fire on any metric or log pattern breach, delivering notifications to the platform notification channels.
FR-OBS-006	M	SDE-level metrics shall be computed by the Data Platform: resource utilisation, active solutions count, build frequency, and test pass rate.
FR-OBS-007	S	Dashboards shall be role-accessible from the portal: infrastructure view (PA), application view (Developer/TL), quality view (QA/RM), and security view (SE).
FR-OBS-008	S	All platform events (Kafka messages) shall include trace ID for end-to-end correlation from API entry to event emission.
FR-OBS-009	C	The platform shall support synthetic monitoring: configurable scheduled probes verifying service health and correctness from an external perspective.

19 Platform Administration Requirements

Requirements for user management, RBAC, audit trail, identity, and compliance.

ID	Priority	Requirement Statement
FR-ADM-001	M	The platform shall support user account lifecycle management: creation, update, deactivation, role assignment, and deletion.
FR-ADM-002	M	Role-Based Access Control (RBAC) shall govern all platform resource access. Roles shall be assignable at: platform, factory, SDE, and solution levels.
FR-ADM-003	M	Multi-factor authentication (MFA) shall be supported and mandatory for all human users.
FR-ADM-004	M	Single Sign-On (SSO) via OAuth2/OIDC shall be supported, including integration with third-party identity providers.
FR-ADM-005	M	An immutable audit trail shall record every state-modifying action: actor, timestamp, action, resource type, resource ID, before state, after state, result, and IP address.
FR-ADM-006	M	The audit trail shall be append-only; no modification or deletion of audit records shall be permitted.
FR-ADM-007	M	Audit records shall be retained for a minimum of 7 years in regulated environments. Records past hot-storage retention shall be archived to cold storage.
FR-ADM-008	M	Platform administrators shall be able to query the audit trail by: actor, resource, action type, time range, and outcome.
FR-ADM-009	S	Service-to-service communication within the platform perimeter shall use mutual TLS (mTLS).
FR-ADM-010	S	The platform shall generate automated compliance reports for: GDPR, CCPA, ISO 27001, SOC 2 Type II, and SLSA.
FR-ADM-011	S	The platform shall support platform-wide notification preference management: channel (portal, email, SMS, Slack, Teams), frequency, and severity thresholds per user.
FR-ADM-012	C	The platform shall support tenant isolation for multi-tenant deployments: complete data isolation between tenants at storage, network, and API layers.

20 Non-Functional Requirements

Performance, reliability, scalability, security, usability, and compliance requirements that apply across the entire platform.

20.1 Performance

ID	Category	Priority	Requirement Statement
NFR-PERF-001	Performance	M	New SDE provisioning shall complete within 10 minutes for any supported cloud target.
NFR-PERF-002	Performance	M	SDE snapshot shall complete within 60 seconds for a standard SDE definition.
NFR-PERF-003	Performance	M	SDE rollback to a named snapshot shall complete within 5 minutes.
NFR-PERF-004	Performance	M	All API endpoints shall respond within 500ms for 95th percentile requests under normal load.
NFR-PERF-005	Performance	M	The CI pipeline shall begin executing within 30 seconds of receiving a push event trigger.
NFR-PERF-006	Performance	S	The platform shall sustain operations for at least 1,000 concurrent Active SDEs without degradation in provisioning or snapshot times.
NFR-PERF-007	Performance	S	Kafka event processing latency shall not exceed 2 seconds from event publication to consumer receipt under normal load.

20.2 Reliability & Availability

ID	Category	Priority	Requirement Statement
NFR-REL-001	Reliability	M	The platform control plane shall achieve 99.9% monthly uptime (less than 44 minutes downtime per month).
NFR-REL-002	Reliability	M	All production deployments shall be achievable with zero downtime via Blue/Green deployment.
NFR-REL-003	Reliability	M	The scheduled SDE backup success rate shall be 99.9% or higher.
NFR-REL-004	Reliability	M	The platform shall support automated failover for all stateful services with a Recovery Time Objective (RTO) of 15 minutes and Recovery Point Objective (RPO) of 1 hour.
NFR-REL-005	Reliability	S	Critical-path services (API Gateway, Identity, SDE Management, SEM) shall target 99.99% monthly uptime.
NFR-REL-006	Reliability	S	The platform shall gracefully degrade: the unavailability of non-critical services (e.g. AI Agent, Benchmarking) shall not impact core SDE and CI/CD functionality.

20.3 Scalability

ID	Category	Priority	Requirement Statement
NFR-SCALE-001	Scalability	M	All microservices shall support horizontal scaling independently of other services.
NFR-SCALE-002	Scalability	M	The Artefact Registry shall support storage of at least 10 million artefact records without performance degradation.
NFR-SCALE-003	Scalability	S	The platform shall support at least 10,000 concurrent users across all roles without SLA degradation.
NFR-SCALE-004	Scalability	S	The Kafka event bus shall handle a sustained throughput of at least 100,000 events per second across all topics.
NFR-SCALE-005	Scalability	C	The platform shall support global distribution with regional deployments and data residency controls per region.

20.4 Security

ID	Category	Priority	Requirement Statement
NFR-SEC-001	Security	M	All data at rest shall be encrypted using AES-256.
NFR-SEC-002	Security	M	All data in transit shall be encrypted using TLS 1.3 or higher.
NFR-SEC-003	Security	M	Service-to-service communication within the platform perimeter shall use mutual TLS (mTLS).
NFR-SEC-004	Security	M	SDE threat detection to quarantine shall execute within 90 seconds of a confirmed threat signal.
NFR-SEC-005	Security	M	All platform APIs shall enforce authentication and authorisation on every request; unauthenticated requests shall receive a 401 response.
NFR-SEC-006	Security	M	The platform shall undergo an independent penetration test prior to initial production launch and annually thereafter.
NFR-SEC-007	Security	S	API rate limiting shall be enforced at the Gateway to protect against denial-of-service attacks.
NFR-SEC-008	Security	S	All build artefacts shall carry SLSA Level 3 provenance attestations.

20.5 Maintainability & Operability

ID	Category	Priority	Requirement Statement
NFR-MAINT-001	Maintainability	M	All platform services shall be independently deployable without requiring coordinated downtime of other services.
NFR-MAINT-002	Maintainability	M	All services shall expose health check and readiness endpoints conforming to the platform health check standard.

NFR-MAINT-003	Maintainability	M	Database schema migrations shall be backward-compatible and executed without service downtime.
NFR-MAINT-004	Maintainability	S	All platform services shall have a documented runbook covering: startup, shutdown, scaling, backup, restore, and common incident procedures.
NFR-MAINT-005	Maintainability	S	API backward compatibility shall be maintained within a major version. Breaking changes shall require a new major version with a minimum 6-month deprecation notice.

20.6 Usability & Accessibility

ID	Category	Priority	Requirement Statement
NFR-UX-001	Usability	M	The platform portal shall meet WCAG 2.1 Level AA accessibility standards.
NFR-UX-002	Usability	M	New developer onboarding (account creation through first SDE active) shall be completable within 30 minutes.
NFR-UX-003	Usability	S	The portal shall support light and dark display modes.
NFR-UX-004	Usability	S	All CLI tools and IDE plugins shall provide context-sensitive help and autocomplete.
NFR-UX-005	Usability	C	The platform shall support localisation (i18n) for a minimum of 5 languages in the portal UI.

20.7 Compliance

ID	Category	Priority	Requirement Statement
NFR-COMP-001	Compliance	M	The platform shall maintain an immutable audit trail with a minimum 7-year retention period for regulated deployment configurations.
NFR-COMP-002	Compliance	M	The platform shall generate audit evidence supporting SOC 2 Type II certification for the Trust Service Criteria: Security, Availability, Processing Integrity, Confidentiality, and Privacy.
NFR-COMP-003	Compliance	S	The platform shall provide evidence artefacts sufficient for ISO 27001 certification without manual data collection.
NFR-COMP-004	Compliance	S	The platform shall support configurable data residency: ensuring all data for a given tenant or deployment is stored and processed within a specified geographic region.
NFR-COMP-005	Compliance	C	The platform shall support FedRAMP Moderate baseline controls for United States federal government deployments.

21 Requirements Traceability Matrix

Maps each requirement domain to its source artefacts in the Qala SDD v2.0 and Workflows & Use Cases documents.

Domain	Requirement IDs	SDD Section	Workflows & Use Cases
Solution Model	FR-SM-001 to FR-SM-012	SDD §3	UC-APP-01, UC-APP-02
SDE	FR-SDE-001 to FR-SDE-035	SDD §4	UC-SDE-01 to UC-SDE-06
Solution Factory	FR-SF-001 to FR-SF-011	SDD §5	UC-SF-01 to UC-SF-03
Hermetic Build	FR-HB-001 to FR-HB-012	SDD §4	UC-CI-01, UC-CI-02
Continuous Integration	FR-CI-001 to FR-CI-013	SDD §5, §8	UC-CI-01
Release & Deployment	FR-REL-001 to FR-DEP-011	SDD §9	UC-REL-01 to UC-REL-04
Config & CM	FR-CM-001 to FR-CM-014	SDD §7	UC-CM-01, UC-CM-02
Bug Tracking	FR-BUG-001 to FR-BUG-011	SDD §8	UC-BUG-01, UC-BUG-02
Test Management	FR-TEST-001 to FR-PROTO-006	SDD §10	UC-TEST-01 to UC-TEST-03
Tooling	FR-TOOL-001 to FR-TOOL-012	SDD §11	UC-TOOL-01, UC-TOOL-02
Security	FR-SEC-001 to FR-PRIV-008	SDD §13	UC-SEC-01 to UC-SEC-03
Benchmarking	FR-BENCH-001 to FR-BENCH-012	SDD §12	UC-BENCH-01, UC-BENCH-02
AI & Intelligence	FR-AI-001 to FR-AI-013	SDD §10, §14	UC-AI-01 to UC-AI-05
Project Management	FR-PM-001 to FR-PM-010	SDD §14	UC-DEV-01
Artefact Management	FR-ART-001 to FR-ART-012	SDD §16	UC-CI-01, UC-REL-01
Environment Management	FR-ENV-001 to FR-ENV-010	SDD §15	UC-SDE-01, UC-CI-01
Observability	FR-OBS-001 to FR-OBS-009	SDD §11	Cross-cutting
Platform Admin	FR-ADM-001 to FR-ADM-012	SDD §9, §13	Cross-cutting
Non-Functional	NFR-PERF through NFR-COMP	SDD §2, §17	All workflows

21.1 Requirement Count Summary

Domain	Must Have	Should Have	Could Have	Total

Solution Model	8	2	2	12
SDE	20	10	5	35
Solution Factory	7	2	2	11
Hermetic Build	9	2	1	12
Continuous Integration	9	3	1	13
Release & Deployment	14	5	1	20
Config & CM	8	6	0	14
Bug Tracking	7	3	1	11
Test Management	7	5	1	18
Tooling	8	4	1	12
Security	16	10	2	28
Benchmarking	6	5	2	12
AI & Intelligence	6	6	2	13
Project Management	6	4	1	10
Artefact Management	8	4	1	12
Environment Management	7	3	1	10
Observability	6	2	1	9
Platform Admin	8	4	1	12
Non-Functional	28	18	6	52
TOTAL	192	102	33	327

End of Document — Qala Solution Factory OS | Platform Requirements Document v1.0